

Los Angeles Regional Water Quality Control Board

October 3, 2022

Uzi Daniel
West Basin Municipal Water District
17140 South Avalon Blvd.
Carson, CA 90746

REQUEST TO USE ALTERNATE TEST SPECIES FOR CHRONIC TOXICITY TESTING, JUANITA MILLENDER-MCDONALD CARSON REGIONAL WATER RECLAMATION PLANT (ORDER NO. R4-2018-0090, CI-7972, NPDES NO. CA0064246)

Dear Ms. Daniel:

The Los Angeles Regional Water Quality Control Board (Los Angeles Water Board) received an e-mail on August 30, 2022, requesting approval to use an alternate test species to satisfy chronic toxicity testing requirements for the Juanita Millender-McDonald Carson Regional Water Reclamation Plant (Carson WRP). The e-mail included a letter from the testing laboratory, Wood Environment and Infrastructure Solutions, Inc., dated August 1, 2022, detailing the challenges of performing chronic toxicity testing on *Haliotis rufescens* (red abalone) and requesting *Strongylocentrotus purpuratus* (purple urchin) as an alternate test species when testing on red abalone is unsuccessful.

The Carson WRP discharges reverse osmosis brine waste to a surge tower where it mixes with secondary-treated wastewater from Los Angeles County Sanitation District's Joint Water Pollution Control Plant before being released to the Pacific Ocean. Order No. R4-2018-0090 requires annual chronic toxicity testing on the combined effluent using the most sensitive species. Out of the three species specified in the Order to be used for testing, currently the most sensitive species is red abalone. However, because of the red abalone's seasonal nature, the lab cannot consistently spawn red abalone during the late summer to fall seasons affecting their ability to produce embryos for the shell development test. In addition, there is currently only one red abalone culturing farm on the West Coast, which at times has limited production of gravid red abalone capable of spawning. Due to these factors, the need for repurchasing and retesting the species can significantly increase the cost of testing with no guarantee of successful testing.

The Ocean Plan specifies Tier 1 test methods that are the preferred toxicity tests for compliance monitoring. Besides the red abalone development test, the Ocean Plan lists

JAMES STAHL, ACTING CHAIR | RENEE PURDY, EXECUTIVE OFFICER

other invertebrate species tests under the Tier 1 classification, such as the purple urchin development test which is the alternate test requested by the laboratory. The laboratory indicates in their letter that the two test species are almost identical in methodology and in their sensitivity to reference toxicants.

After reviewing the request and regulatory requirements, the Los Angeles Water Board approves the purple urchin development test listed as Tier 1 in the Ocean Plan as an alternate test species for compliance with this year's annual chronic toxicity test if the red abalone test conducted in September 2022 is unsuccessful. As the red abalone is the current most sensitive species, there is no alternate species specified in the permit, and alternative test methods require Executive Officer approval, any future requests to substitute the abalone with the purple urchin must be requested in writing for consideration prior to substitution.

Should you have any questions regarding this matter, please contact Danielle Robinson at Danielle.Robinson@waterboards.ca.gov or Steven Webb at Steven.Webb@waterboards.ca.gov.

Sincerely,

Renee Purdy
Executive Officer

cc: Jamie Malpede, West Basin Municipal Water District